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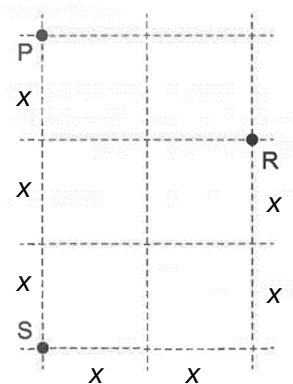
B

Distance SR

$$= \sqrt{(2x)^2 + (2x)^2} = 2.828x$$

Distance SP

$$= 3x$$



$$\text{Since, } V = \frac{Q}{4\pi\epsilon_0 r}, V \propto \frac{1}{r},$$

$$\frac{V_P}{V_R} = \frac{2.828x}{3x}$$

$$\frac{V_P}{636} = \frac{2.828}{3}$$

$$V_P = 599.5$$

$$\approx 600 \text{ V}$$

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B

The e m f. of a cell is the amount of energy converted into electrical energy